

Multiple Choice (10 marks)

1. A satellite of mass m orbits a planet of mass M in a circular orbit of radius R . The time required for one revolution is
 - (a) independent of M
 - (b) proportional to $m^{1/2}$
 - (c) linear in R
 - (d) proportional to $R^{3/2}$
2. The energy required to remove both electrons from the helium atom in its ground state is 79.0 eV. How much energy is required to ionize helium (i.e., to remove one electron)?
 - (a) 24.6 eV
 - (b) 39.5 eV
 - (c) 51.8 eV
 - (d) 54.4 eV
3. A grating spectrometer can just barely resolve two wavelengths of 500 nm and 502 nm, respectively. Which of the following gives the resolving power of the spectrometer?
 - (a) 2
 - (b) 250
 - (c) 5,000
 - (d) 10,000
4. The eigenvalues of a Hermitian operator are always
 - (a) real
 - (b) imaginary
 - (c) degenerate
 - (d) linear
5. The negative muon, μ^- , has properties most similar to which of the following?
 6. Electron
 7. Meson
 8. Photon
 9. Boson

True / False (10 marks)

Answer True or False

6. True or False: Electromagnetic radiation emitted from a nucleus is often observed as visible light.
7. True or False: Electrons in metals possess fewer degrees of freedom than the atoms in the lattice, resulting in lower kinetic energy.
8. True or False: A helium-neon laser is the best type of laser for spectroscopy over a range of visible wavelengths.
9. True or False: The angular magnification of the refracting telescope can be calculated as 5 with the given focal lengths.
10. True or False: An observer must move at 0.60 c parallel to a rod measuring 1.00 m in its rest frame to measure its length as 0.80 m .

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the motion of a payload dropped from an airplane traveling north at 100 m/s , analyzing its velocity relative to the plane 4.0 seconds after release, assuming no air resistance.
12. Discuss the implications of a reversible thermodynamic process on the entropy of a system and its environment, and analyze how this reflects the principles of thermodynamics.

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